MILITARY SPECIFICATION SHEET

MULTIPLEXERS, DEMULTIPLEXERS, MULTIPLEXERS/DEMULTIPLEXERS (MULDEMS), DEMULTIPLEXER, FREQUENCY-DIVISION, FIBER OPTIC, SHIPBOARD, 1:4 CHANNELS, 30 MHZ (DOUBLE SIDEBAND) (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24731.

CLASSIFICATION

Equipment function: D (Frequency-division demultiplexer).
Signal type: B (Optical).
Channel configuration: A (1 to 16 channels).
Analog subchannel class: F (Not greater than 30 MHz).
Digital subchannel class: G (Not greater than 100 Mbps).
Style: C (Vertical rack plug-in module).

RELIABILITY: MTBF greater than 10,000 hours.

DESIGN AND CONSTRUCTION

Enclosure: Watertight.
Fiber optic transmitters: N/A.
Fiber optic receivers: The receiver shall operate at 1.30 μm with multimode fiber. The maximum receiver input power shall be 24 dBmW.
Fiber optic pigtail: In accordance with MIL-C-85045 and MIL-C-85045/14.
Fiber optic connectors: N/A.

Dimensions:

Mainframe assembly: See figure 1.
Demultiplexer module: See figure 2.
Fiber optic receiver module: See figure 3.
Power supply module: See figure 4.

AMSC N/A
FSC 6008
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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is not mandatory. may be helpful. PUT

Weight:

Mainframe assembly with power supply: 6.8 kg (max.).
Demultiplexer module: 1.1 kg (max.).
Fiber optic receiver module: 0.7 kg (max.).

GENERAL PERFORMANCE CHARACTERISTICS

Demultiplexers: 1 input channel, 4 output channels; 70 MHz per output channel.

Subchannel output interface element: The subchannel interface element shall accept a frequency range from 52.5 to 577.5 MHz in 35 MHz steps and demodulate to 4 channels with an intermediate frequency (IF) of 70 MHz.

Separator/amplifier element: Passive 4-way separator.

Demultiplexer input interface element: The demultiplexer input interface element output shall accept an optically modulated input signal in accordance with MIL-R-AAAAA. The electrical output of the fiber optic receiver shall have a frequency range from 52.5 to 577.5 MHz in 35 MHz steps with a 3 dB double-sideband bandwidth of 30 MHz.

DETAILED PERFORMANCE CHARACTERISTICS

Operational performance parameters:

Prime power and current load for demultiplexer:

5 ± 0.25 volts	at 240 mA max.
15, minus 0.75, plus 2.25 volts	at 300 mA max.
Minus 15, minus 0.75, plus 2.25 volts	at 35 mA max.

Prime power and current load for receiver unit:

15, minus 0.75, plus 2.25 volts	at 400 mA max.
Minus 15, minus 0.75, plus 2.25 volts	at 40 mA max.
5, minus 0.25, plus 0.75 volts	at 30 mA max.

Ac prime power: 120 volts, 60 Hz, 7.5 amperes max.

Alarm and fault conditions: Status monitoring for power supplies, input signal voltage level, and receiver output levels.

Alarm indication signal: N/A.

Demodulation formats: Frequency demodulation in accordance with the frequency plan.

Guard-band width: 5 MHz.

Propagation delay: 10 nanoseconds.

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Pilot tone characteristics: N/A.

Frequency plan: 52.5 to 577.5 MHz in 35 MHz steps.

Interchannel crosstalk: minus 40 dB (less than or equal to).

Optical performance:
Dynamic range: 20 dB.

Channel performance parameters:

Analog subchannels: The IF output signals of the multiplexer module shall have the following characteristics:

Impedance: 75 ohms.
Return loss: 20 dB, minimum.
Frequency: 70 MHz.
Power level: minus 3 dBmW ± 1 dB.
Frequency stability: plus or minus 50 kHz.
Frequency accuracy: plus or minus 50 kHz.

Digital channels: N/A.
Other signal types: N/A.

Interface characteristics:

Analog subchannel interface: The RF input signals shall have the following characteristics:

Impedance: 75 ohms.
Return loss: 15 dB, minimum.
Frequency range: 52.5 to 577.5 MHz.
Power level: minus 33 to minus 23 dBmW.

Digital channels: N/A.

Multiplexed signal interface:

FDM signal output: N/A.
FDDM signal input:

Electrical interface: N/A.
Optical interface: In accordance with input signals of MIL-R-24737.

Maintenance orderwire: N/A.
Other interfaces: N/A.

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ENVIRONMENTAL REQUIREMENTS

Temperature range: 1 (operating - 0 to plus 85°C;
storage - minus 62 to plus 85°C).

Nuclear radiation resistance: N/A:

ELECTROMAGNETIC INTERFERENCE SUPPRESSION

Electromagnetic: Below-deck application, in accordance with MIL-STD-461, part 5.

Conducted susceptibility: CS01, CS02, CS06, CS09.
Conducted emission: CE01, CE03, CE06.
Radiated emission: RE01, RE02, RE03.
Radiated susceptibility: RS01, RS02, RS03.

Lightning: Transient suppression for voltage spikes of 2500 volts.
Power leads: N/A.

SAFETY

Laser safety: N/A.

METHODS OF INSPECTION

OPERATIONAL TESTS

Pilot tones: N/A.

ENVIRONMENTAL TESTS

Explosive atmosphere: The values for the test conditions shall be
Fuel - n-hexane (reagent grade or 95 percent).
Fuel amount - 3.8 percent by volume of the test chamber.
Test altitude - Sea level.
Test temperature - Plus 85°C.

Electromagnetic interference suppression:

Tests shall be conducted in accordance with MIL-STD-462, over a frequency range of 14 KHz to 600 MHz. The RS03 limit shall be 200 volts per meter. Tests shall be conducted at ambient conditions. Verification of operation shall be performed after the test.

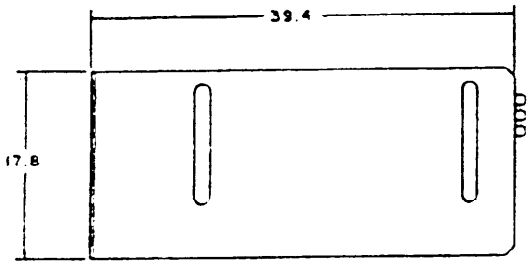
PART OR IDENTIFYING NUMBER: D24731/02-001.

PART DESIGNATOR: DBAFGC.

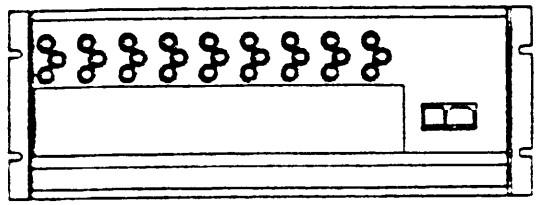
Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
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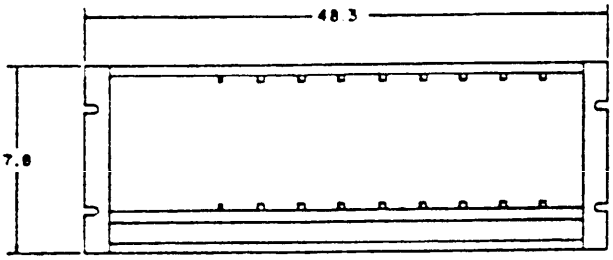
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SIDE VIEW



BACK VIEW

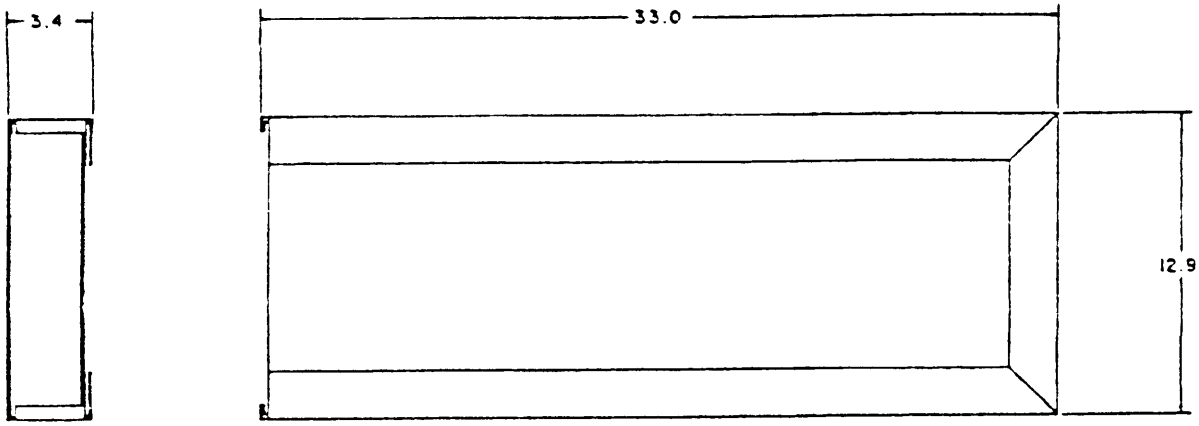


FRONT VIEW

NOTE:
1. All dimensions are in centimeters.

FIGURE 1. Mainframe assembly.

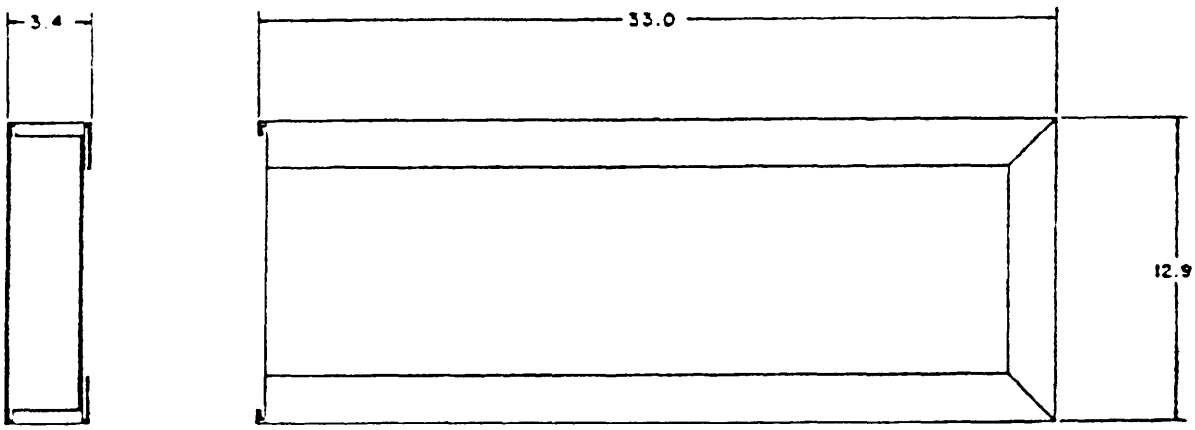
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NOTE:
1. All dimensions are in centimeters.

FIGURE 2. Demultiplexer module.

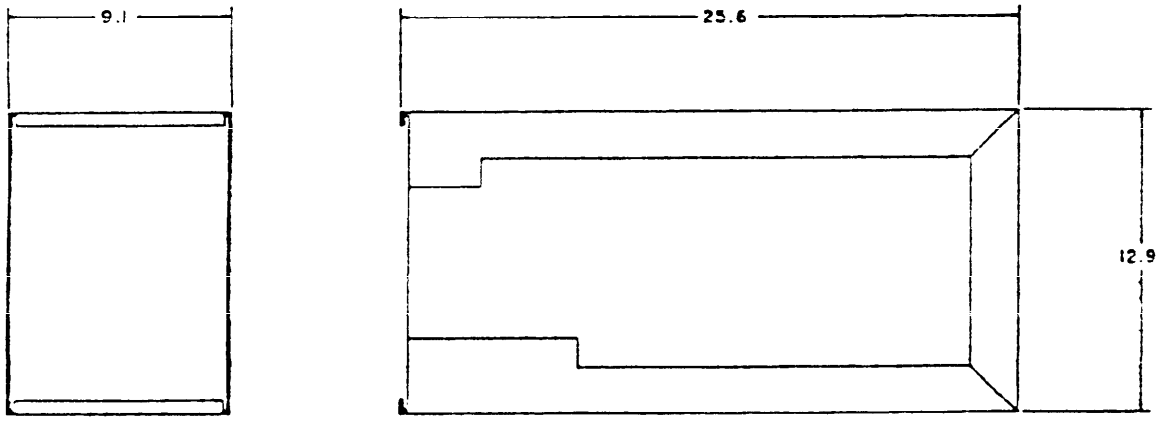
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NOTE:
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FIGURE 3. Fiber optic receiver module.

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NOTE:
1. All dimensions are in centimeters.

FIGURE 4. Power supply module.

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